

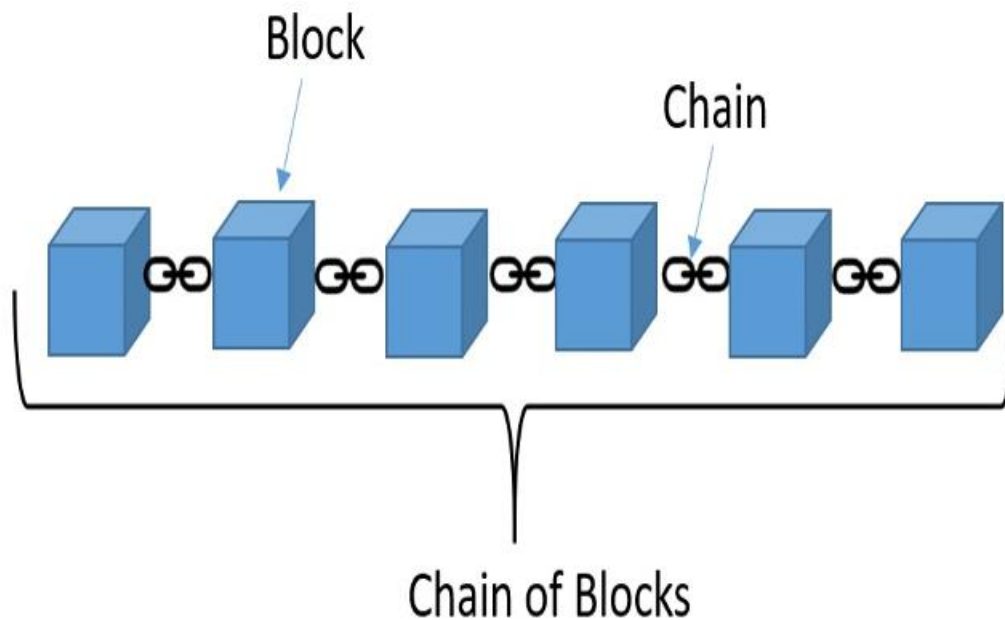
What is a blockchain?

A blockchain is essentially a decentralized distributed database or a ledger, as follows:

- **Decentralization:** In simple terms, it means that the application or service continues to be available and usable even if a server or a group of servers on a network crashes or is not available. The service or application is deployed on a network in a way that no server has absolute control over data and execution, rather each server has a current copy of data and execution logic.
- **Distributed:** This means that any server or node on a network is connected to every other node on the network. Rather than having one-to-one or one-to-many connectivity between servers, servers have many-to-many connections with other servers.
- **Database:** This refers to the location for storing durable data that can be accessed at any point in time. A database allows storage and retrieval of data as functionality and also provides management functionalities to manage data efficiently, such as export, import, backup, and restoration.
- **Ledger:** This is an accounting term. Think of it as specialized storage and retrieval of data. Think of ledgers that are available to banks. For example, when a transaction is executed with a bank—say, Tom deposits 100 dollars in his account, the bank enters this information in a ledger as a credit. At some point in the future Tom withdraws 25 dollars. The bank does not modify the existing entry and stored data from 100 to 75. Instead it adds another entry in the same ledger as a debit of 25 dollars. It means a ledger is a specialized database that does not allow modification of existing data. It allows you to create and append a new transaction to modify the current balance in the ledger. The blockchain is a database that has the same characteristics of a ledger. It allows newer transactions to be stored in an append-only pattern without any scope to modify past transactions. It is important here to understand that existing data can be modified by using a new transaction, but past transactions cannot be modified. A balance of 100 dollars can be modified at any time by executing a new debit or credit transaction, but previous transactions cannot be modified.

Note (from me): we can implement ledger for each customer, where the nonce value depends on the previous transactions, then store them in immutable database.

Note (From me): in the previous way, by using blockchain we can show the history of transactions in our system in safe way, and we can track the data in our financial system



Blockchain means a chain of blocks. Blockchain means having multiple blocks chained together, with each block storing transactions in a way where it is not possible to change these transactions.

Not being able to change and modify past transactions makes blockchain solutions highly trustworthy, transparent, and incorruptible.

Why blockchains?

The main objective of Ethereum is to accept transactions from accounts, update their state, and maintain this state as current till another transaction updates it again. The whole process of accepting, executing, and writing transactions can be divided into two phases in Ethereum. There is a decoupling between when a transaction is accepted by Ethereum and when the transaction is executed and written to the ledger. This decoupling is quite important for decentralization and distributed architecture to work as expected.

Blockchain helps primarily in the following three different ways:

- **Trust:** Blockchain helps in creating applications that are decentralized and collectively owned by multiple people. Nobody within this group has the power to change or delete previous transactions. Even if someone tries to do so, it will not be accepted by other stakeholders.
- **Autonomy:** There is no single owner for blockchain-based applications. No one controls the blockchain, but everyone participates in its activities. This helps in creating solutions that cannot be manipulated or induce corruption.
- **Intermediaries:** Blockchain-based applications can help remove the intermediaries from existing processes. Generally, there is a central body, such as vehicle registration, license issuing, and so on, that acts as registrar for registering vehicles as well as issuing driver licenses.

Without blockchain-based systems, there is no central body and if a license is issued or vehicle is registered after a blockchain mining process, that will remain a fact for an epoch time-period without the need of any central authority vouching for it.

Ethereum uses the hashing technique quite extensively. It hashes every transaction, hashes the hash of two transactions at a time, and ultimately generates a single root transaction hash for every transaction within a block.

Ether

Ether is the currency of Ethereum. Every activity on Ethereum that modifies its state costs Ether as a fee, and miners who are successful in generating and writing a block in a chain are also rewarded Ether. Ether can easily be converted to dollars or other traditional currencies through *crypto-exchanges*. Ethereum has a metric system of denominations used as units of Ether. The smallest denomination or base unit of Ether is called *wei*.

Gas

Fees are paid using Ether for any execution that *changes state in Ethereum*. Ether is traded on public exchanges and its price fluctuates daily. If Ether is used for paying fees, then the cost of using the same service could be very high on certain days and low on other days. People will wait for the price of Ether to fall to execute their transactions. This is not ideal for a platform such as Ethereum. Gas helps in alleviating this problem. Gas is the internal currency of Ethereum. The execution and resource utilization cost is predetermined in Ethereum in terms of gas units. This is also known as gas cost. There is also gas price that can be adjusted to a lower price when the price of Ether increases and a higher price when the price of Ether decreases. For example, to invoke a function in a contract that modifies a string will cost gas, which is predetermined, and users should pay in terms of *gas to ensure smooth execution of this transaction*.

A blockchain network *consists of multiple nodes* belonging to miners and some nodes that do not mine but help in execution of smart contracts and transactions. *These are known as EVMs*. Each node is connected to another node on the network. These nodes use peer-to-peer protocol to talk to each other. They, by default, use port 30303 to talk among themselves.

Each miner maintains an instance of ledger. A ledger contains all blocks in the chain. With multiple miners it is quite possible that each miner's ledger instance might have different blocks to another. The miners synchronize their blocks on an on-going basis to ensure that every miner's ledger instance is the same as the other.

The EVM also hosts smart contracts. Smart contracts help in extending Ethereum by writing custom business functionality into it. These smart contracts can be executed as part of a transaction and it follows the process of mining (From above).

A person having an account on a network can send a message for transfer of Ether from their account to another or can send a message to invoke a function within a contract. Ethereum does not distinguish them as far as transactions are considered. The transaction must be digitally signed with an account holder's private key. This is to ensure that the identity of the sender can be established while verifying the transaction and changing the balances of multiple accounts.

How are blocks related to each other?

In blockchain and Ethereum every block is related to another block. There is a parent-child relationship between two blocks. There can be only one child to a parent and a child can have a single parent. This helps in forming a chain in blockchain. Blocks are explained in a later section in this chapter. In the following diagram, three blocks are shown—**Block 1**, **Block 2**, and **Block 3**. **Block 1** is the parent of **Block 2** and **Block 2** is the parent of **Block 3**. The relationship is established by storing the parent block's hash in a child's block header. **Block 2** stores the hash of **Block 1** in its header and **Block 3** stored the hash of **Block 2** in its header. So, the question arises—who is the parent of the first block? Ethereum has a concept of the *genesis block* also known as **first block**. This block is created automatically when the chain is first initiated. You can say that a chain is initiated with the first block known as the Genesis Block and the formation of this block is driven through the **genesis.json** file.

How are transactions and blocks related to each other?

Each block has an upper gas limit and each transaction needs a certain amount of gas to be consumed as part of its execution. The cumulative gas from all transactions that are not yet written in a ledger cannot surpass the block gas limit. This ensures that all transactions do not get stored within a single block. As soon as the gas limit is reached, other transactions are removed from the block and mining begins thereafter.

Note (From me): depending on above we can write financial systems that has limit constraints for each operation, so no in-valid record will be exists

The transactions are hashed and stored in the block. The hashes of two transactions are taken and hashed further to generate another hash. This process eventually provides a single hash from all transactions stored within the block. This hash is known as the transaction Merkle root hash and is stored in a block's header. A change in any transaction will result in a change in its hash and, eventually, a change in the root transaction hash. It will have a cumulative effect because the hash of the block will change, and the child block has to change its hash because it stores its parent hash. This helps in making transactions immutable.

There are the following two types of nodes in Ethereum:

- EVM
- Mining nodes

EVM:

Think of EVM as the **execution runtime** for **an Ethereum network**. EVMs are primarily responsible for *providing a runtime that can execute code written in smart contracts*. It can access accounts, both contract and externally owned, and its own storage data. It does not have access to the overall ledger but does have limited information about the current transaction.

Ethereum mining nodes

A miner is responsible for writing transactions to the Ethereum chain. A miner's job is very similar to that of an accountant. *As an accountant is responsible for writing and maintaining the ledger;* similarly, a miner is solely responsible for writing a transaction to an Ethereum ledger. A miner is interested in writing transactions to a ledger because of the reward associated with it. Miners get two types of reward—

1. a reward for writing a block to the chain and,
2. cumulative *gas fees* from all *transactions in the block*.

There are generally many miners available within a blockchain network each trying and competing to write transactions. However, only one miner can write the block to the ledger and the rest will not be able to write the current block.

The next task for a miner is to generate the block header and perform the following tasks:

1. The miner **takes hashes of two transactions at a time to generate** a new hash till he gets a single hash from all transactions. The hash is referred to as a root transaction **hash or Merkle root transaction hash**. This hash is added to the block header.
2. The miner also identifies the hash of the previous block. The previous block *will become parent to the current block* and its hash will also be added to the block header.
3. The miner calculates the state and **receipts of transaction root hashes** and adds them to the block header.
4. *A nonce and timestamp is also added* to the block header.
5. A block hash consisting of both block header and body is generated.
6. The mining process starts where the miner keeps changing the nonce value and tries to find a hash that will **satisfy as an answer to the given puzzle**. It is to be kept in mind that everything mentioned here is executed by every miner in the network.
7. Eventually, one of the miners will be able to solve the puzzle and advertise the same to other miners in the network. The other miners will verify the answer and, if found correct, will further **verify every transaction, accept the block, and append the same to their ledger instance**.

This entire process is also known as **Proof of Work (PoW)** wherein a miner provides proof that it is has worked on computing the final answer that could satisfy as solution to the puzzle. There are other algorithms such as **Proof of Stake (PoS)** and **Proof of Authority (PoA)**

